

First-order theories

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1 First order theories

Any *first order theory* T is defined as FOL having some distinguished predicate, function or constant symbols in the alphabet, along with special axioms involving these symbols (apart from all the FOL-axioms). The distinguished symbols are termed the *proper symbols* of T , and the special axioms, which are closed wffs, are called the *proper axioms* of T .

Let us note the following.

- In an abuse of notation, the set of proper axioms of T will also be denoted by T .
- When we say α is a wff of T , we mean that all the proper symbols occurring in α (if any) come from the set of proper symbols of T .
- $\mathcal{I}$ is an interpretation or a model of T means that $\mathcal{I}$ is an interpretation or a model for all the proper axioms of T .
- α is a theorem of T , sometimes written as $\vdash_T \alpha$, means that $T \vdash \alpha$, i.e. α is deduced (through the standard FOL deduction procedure) from the proper axioms of T and the FOL-axioms. This would imply that, in particular, all the FOL-theorems are also T -theorems.
- All the results about any set Γ of closed wffs in FOL, apply to all first order theories T with countable language, considering T to also denote the set of proper axioms (which are closed wffs) of the theory. So we

would have, e.g. if T is consistent, it has a (countable) model, or if $T \models \alpha$ then $T \vdash \alpha$, for any closed wff α in T .

2 Equality

A *first-order theory with equality* has in its language the special binary predicate symbol $=$, and as its axioms (i) the *FOL* axioms, (ii) the *equality axioms*, and (iii) a set of *proper axioms*.

The equality axioms:

$$\text{E1 } \forall x(x = x).$$

$$\text{E2 } \forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n (x_1 = y_1 \wedge \dots \wedge x_n = y_n \rightarrow (p^n x_1 \dots x_n \leftrightarrow p^n y_1 \dots y_n)).$$

$$\text{E3 } \forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n (x_1 = y_1 \wedge \dots \wedge x_n = y_n \rightarrow (f^n x_1 \dots x_n = f^n y_1 \dots y_n)).$$

Using the axioms, one can prove symmetry and transitivity of $=$, i.e.

1. $\forall x \forall y (x = y \rightarrow y = x)$, and
2. $\forall x \forall y \forall z (x = y \wedge y = z \rightarrow x = z)$.

2.1 Some examples of first order theories with equality

We just specify the set $\mathcal{P}$ of proper symbols, and the proper axioms.

(i) **Groups**

$\mathcal{P} := \{=, +, 0\}$, $+$: binary function symbol and 0 : constant;

Proper axioms:

$$\forall x \forall y \forall z ((x + y) + z = x + (y + z))$$

$$\forall x (x + 0 = x)$$

$$\forall x \exists y (x + y = 0)$$

Observe that the equality axiom E3 applied on $+$ takes the form:

$$\forall x_1 \forall x_2 \forall y_1 \forall y_2 ((x_1 = y_1 \wedge x_2 = y_2) \rightarrow (x_1 + x_2 = y_1 + y_2)).$$

- (ii) **Abelian Groups:** Groups with the additional proper axiom
 $\forall x \forall y (x + y = y + x)$
- (iii) **Partially Ordered Sets**
 $\mathcal{P} := \{=, \leq\}$, $\leq$: binary predicate symbol;
Proper axioms:
 $\forall x (x \leq x)$
 $\forall x \forall y ((x \leq y \wedge y \leq x) \rightarrow x = y)$
 $\forall x \forall y \forall z ((x \leq y \wedge y \leq z) \rightarrow x \leq z)$
- (iv) **Lattices**
 $\mathcal{P} := \{=, \leq, \cup, \cap\}$, $\leq$: binary predicate symbol, $\cup, \cap$: binary function symbols;
Proper axioms:
 Proper axioms for Partially Ordered Sets;
 $\forall x \forall y (x \leq x \cup y \wedge y \leq x \cup y)$
 $\forall x \forall y (x \cap y \leq x \wedge x \cap y \leq y)$
 $\forall x \forall y \forall z ((x \leq z \wedge y \leq z) \rightarrow x \cup y \leq z)$
 $\forall x \forall y \forall z ((z \leq x \wedge z \leq y) \rightarrow z \leq x \cap y)$
- (v) **Dense Linearly Ordered Sets without End-Points**
 $\mathcal{P} := \{=, <\}$, $<$: binary predicate symbol;
Proper axioms:
 $\forall x \neg (x < x)$
 $\forall x \forall y (x < y \vee y < x \vee x = y)$
 $\forall x \forall y \forall z ((x < y \wedge y < z) \rightarrow x < z)$
 $\forall x \forall y (x < y \rightarrow \exists z (x < z \wedge z < y))$
 $\forall x \exists y (y < x)$
 $\forall x \exists y (x < y)$
- (vi) **Number Theory**
 $\mathcal{P} := \{=, ', +, \times, 0\}$, $+, \times$: binary function symbols, $'$: unary function symbol (interpreted as the ‘successor’ function);
Proper axioms:
 $\forall x \forall y (x' = y' \rightarrow x = y)$
 $\forall x (x' \neq 0)$
 $\forall x (x + 0 = x)$
 $\forall x \forall y (x + y' = (x + y)')$
 $\forall x (x \times 0 = 0)$
 $\forall x \forall y (x \times y' = (x \times y) + x)$
 For all wffs $\alpha(x)$ in the language, the closure of
 $(\alpha(0/x) \wedge \forall x (\alpha(x) \rightarrow \alpha(x'/x))) \rightarrow \forall x \alpha(x)$

Exercise 2.1.

- (i) Formulate the first order theories of (a) Rings, (b) Fields, and (c) Boolean Algebras.
- (ii) Consider Number Theory, as defined above. Can you show, informally, that no model of Number Theory can be finite?

3 Uses of Equality predicate in FOL language

3.1 Uniqueness

Without equality, we cannot express the notion of *uniqueness*. We use the notation $\exists!x\alpha(x)$ to denote “there exists a unique x such that $\alpha(x)$ holds”. Formally, $\exists!x\alpha(x)$ is an abbreviation of the wff

$$\exists x\alpha(x) \wedge \forall x\forall y((\alpha(x) \wedge \alpha(y)) \rightarrow x = y),$$

where y is the first variable in the enumeration of variables that is not in $\alpha(x)$.

In fact, $\exists!x\alpha(x)$ defined as above, is logically equivalent to

$$(i) \exists x(\alpha(x) \wedge \forall y(\alpha(y) \rightarrow y = x)),$$

and also to

$$(ii) \exists x\forall y(x = y \leftrightarrow \alpha(y)).$$

It is a non-trivial exercise to prove the equivalence, but you can try (syntactically, or semantically, as you wish).

An easy consequence:

$$\vdash \forall x\exists!y(y = x).$$

A proof: Note that

$$\exists!y(y = x) := \exists y(y = x) \wedge \forall y\forall z((y = x \wedge z = x) \rightarrow y = z).$$

To prove the first part of the conjunct, use reflexivity ($x = x$), followed by Rule $\exists$. For the other part, the properties of symmetry and transitivity of equality proved earlier, give $(y = x \wedge z = x) \rightarrow y = z$. Two uses of Rule Gen give $\forall y \forall z ((y = x \wedge z = x) \rightarrow y = z)$.

3.2 At least, at most, exactly

We can express notions such as “at least” or “at most”, with the help of equality.

Example 3.1.

- (i) The wff $\exists x \exists y (\alpha(x) \wedge \alpha(y) \wedge x \neq y)$ expresses that “there exist at least two x ’s such that $\alpha(x)$ ”. More simply, the wff $\exists x \exists y (x \neq y)$ expresses that there exist at least two elements. In other words, for any interpretation $\mathcal{I}$ of a first order theory T with equality,

$\mathcal{I} \models \exists x \exists y (x \neq y)$, if and only if $\mathcal{I}$ has at least two elements.

In general, for $n \geq 2$, we denote by ‘ $\exists n$ ’ the wff

$$\exists x_1 \dots \exists x_n (x_1 \neq x_2 \wedge x_1 \neq x_3 \wedge \dots \wedge x_1 \neq x_n \wedge x_2 \neq x_3 \wedge \dots \wedge x_{n-1} \neq x_n),$$

giving the following.

- (a) For any interpretation $\mathcal{I}$ of T ,

$$\mathcal{I} \models \exists n, \text{ if and only if } |D| \geq n,$$

where $|D|$ denotes the cardinality of the domain D of $\mathcal{I}$.

- (b) If $n \geq k$, $\vdash \exists n \rightarrow \exists k$.

- (ii) Let us enhance the wff $\exists n$, to express the existence of *exactly* n elements. Define the wff ‘ $\exists! n$ ’ as follows.

$$\exists x_1 \dots \exists x_n ((x_1 \neq x_2 \wedge x_1 \neq x_3 \wedge \dots \wedge x_1 \neq x_n \wedge x_2 \neq x_3 \wedge \dots \wedge x_{n-1} \neq x_n) \wedge \forall y (y = x_1 \vee y = x_2 \vee \dots \vee y = x_n)).$$

Here we would have, for any interpretation $\mathcal{I}$ of T ,

$$\mathcal{I} \models \exists! n, \text{ if and only if } |D| = n.$$

4 Some elementary model theory

In this section, we examine some notions and results related to models of first order theories.

As observed for the Completeness theorem, any consistent set of closed FOL-wffs has a model with a countable domain. So, using the fact that any set having a model is consistent, we obtain the first result in this section.

Theorem 4.1. (Löwenheim-Skolem – downwards) If T has a model, it has a countable model.

There is an ‘upwards’ version of the theorem:

Theorem 4.2. (Löwenheim-Skolem – upwards) If T has an infinite model, then for any transfinite cardinal number κ , T has a model with cardinality exactly κ .

There is another fundamental result that we obtain very simply, as a consequence of the completeness theorem.

Theorem 4.3. (Compactness) If every finite subset of T has a model, T itself has a model.

Proof. Argue by contradiction: if T does not have a model, by the Completeness theorem, T is inconsistent, i.e. $T \vdash \square$. Then, by compactness of ‘ $\vdash$ ’, there is a finite subset T_0 of T which is inconsistent, and hence does not have a model. $\square$

Let us illustrate a use of the compactness theorem.

Proposition 4.4. Let T be a first order theory with equality. Suppose T has a model with cardinality at least n , for all positive integers n . Then T has an infinite model.

Proof. Define $T' := T \cup \{\exists n : n = 2, 3, \dots\}$. We apply Theorem 4.3 on T' . Consider any finite subset T_0 of T' . Finiteness of T_0 gives a positive integer n that is the largest amongst all m such that the wff $\exists m$ belongs to T_0 . By assumption, there is a model $\mathcal{I}$ of T with at least n elements. Using (a) and (b) of Example 3.1, we get that $\mathcal{I} \models \exists m$, for all m with $m \leq n$. As $\mathcal{I} \models T$, $\mathcal{I}$ is therefore a model of T_0 . Hence by compactness, T' has a model. $\square$

4.1 Completeness of a first order theory

This is a *property* of a first order theory that we shall define – as opposed to the meta-theorem we proved *about* first order theories, viz. the ‘completeness theorem’.

Definition 4.1. A first order theory T is said to be *complete*, provided, for all closed wffs α of T , either $T \vdash \alpha$ or $T \vdash \neg\alpha$.

An obvious consequence: if T is inconsistent, it is complete.

A useful characterization of this notion is given by the following.

Proposition 4.5. T is complete, if and only if every closed wff of T that is true in one model of T is true in all models of T .

Proof. ($\Rightarrow$) Suppose $\mathcal{I} \models \alpha$, where $\mathcal{I}$ is a model of T . As T is complete, either $T \vdash \alpha$ or $T \vdash \neg\alpha$. But the latter cannot be the case, as $\mathcal{I} \models \alpha$. Then $T \vdash \alpha$ implies that $\mathcal{I}' \models \alpha$, for all models $\mathcal{I}'$ of T .

($\Leftarrow$) Divide into two cases: (i) T is inconsistent, or (ii) T is consistent.

(i) trivially gives completeness, as noted above.

(ii) T has a model, say $\mathcal{I}$, by the Completeness theorem. For any closed α in T , either $\mathcal{I} \models \alpha$ or $\mathcal{I} \models \neg\alpha$. Using the assumption and the Completeness theorem again, the first leads to $T \vdash \alpha$, and the second to $T \vdash \neg\alpha$. $\square$

Using Proposition 4.5, we can easily produce examples of *incomplete* first order theories. For instance, the theory AG of Abelian groups (cf. (ii) in Section 2.1) is incomplete, as the closed wff $\forall x\forall y(x = y)$ is true in singleton models of AG, but false in all others. Clearly, this argument would apply to all theories having models of different cardinalities, making them incomplete. We shall see examples of complete theories in the next sections.

Recall Number Theory, denoted $\aleph$, formulated as a first order theory in (vi) of Section 2.1. We end this section stating two fundamental results about $\aleph$, due to Kurt Gödel. One may refer to Margaris, for a sketch of the proofs.

Theorem 4.6. (Gödel’s incompleteness theorem) If $\aleph$ is consistent, it is incomplete.

The second theorem (stated informally): the consistency of $\aleph$ cannot be proved in $\aleph$.

The second result is a consequence of the first.

These two results are particularly significant in that they can be applied to theories containing $\aleph$, or those containing theories even weaker than $\aleph$.

4.2 Isomorphism of models

Interpretations of any first order theory T may be regarded as algebraic structures, being tuples consisting of a non-empty domain D along with relations and operations of various arities on D . Given that, it is only natural to discuss correspondences between interpretations of T – in particular, to define the notion of *isomorphism* between models of T .

Definition 4.2. Consider two models $\mathcal{I}$ and $\mathcal{I}'$ of a first order theory T . $\mathcal{I}$ is said to be *isomorphic* to $\mathcal{I}'$, written $\mathcal{I} \cong \mathcal{I}'$, provided the following hold.

- (i) There is a bijection h between D and D' , the domains of $\mathcal{I}$ and $\mathcal{I}'$ respectively.
- (ii) For interpretations R and R' of any n -ary predicate symbol p^n in $\mathcal{I}$ and $\mathcal{I}'$ respectively, and $d_1, \dots, d_n \in D$,
 $(d_1, \dots, d_n) \in R$, if and only if $(h(d_1), \dots, h(d_n)) \in R'$.
- (iii) For interpretations F and F' of any n -ary function symbol f^n in $\mathcal{I}$ and $\mathcal{I}'$ respectively, and $d_1, \dots, d_n \in D$,
 $h(F(d_1, \dots, d_n)) = F'((h(d_1), \dots, h(d_n)))$.
- (iv) For interpretations d and d' of any constant c in $\mathcal{I}$ and $\mathcal{I}'$ respectively, $h(d) = d'$.

For algebraic structures, isomorphisms ‘preserve’ the operations in the structures. Here the property to be preserved is satisfaction of wffs of T . More precisely, we have

Theorem 4.7. Consider two isomorphic models $\mathcal{I}$ and $\mathcal{I}'$ of T , with bijection $h : D \rightarrow D'$. For any wff α of T and valuation v in $\mathcal{I}$,
 $\mathcal{I}, v \models \alpha$, if and only if $\mathcal{I}', h \circ v \models \alpha$.

Proof. By induction on the number of connectives and quantifiers in α . Basis and Boolean cases are easy. Suppose $\alpha := \forall x\beta$.

($\Rightarrow$): Take v' in $\mathcal{I}'$, x -equivalent to $h \circ v$. Define v'' in $\mathcal{I}$, x -equivalent to v such that $v''(x) := (h^{-1} \circ v')(x)$. Then $v' = h \circ v''$, and use induction hypothesis on β . $\square$

Corollary 4.8. $\mathcal{I} \models \alpha$, if and only if $\mathcal{I}' \models \alpha$, with $\mathcal{I}, \mathcal{I}', \alpha$ as in the theorem.

Proof. Exercise! $\square$

The following now provides an example of a complete theory. Consider the first order theory DL of Dense Linearly Ordered Sets without End-Points, defined in (v) of Section 2.1.

Proposition 4.9. DL is complete.

Proof. One can see from the defining axioms of DL that any model of DL must be infinite. To establish the result, we shall assume a result due to Cantor: *all countably infinite models of DL are isomorphic to $\mathcal{Q}$* , the set of rational numbers. Now consider any wff α of DL, and the theories $T_1 := \text{DL} \cup \{\alpha\}$, and $T_2 := \text{DL} \cup \{\neg\alpha\}$.

Claim. One of T_1, T_2 is inconsistent.

The Claim is proved by contradiction. If possible, suppose both are consistent. By the Completeness theorem, there are countable models $\mathcal{I}$ and $\mathcal{I}'$ respectively of T_1 and T_2 , and $\mathcal{I}, \mathcal{I}'$ must be infinite. By Cantor's result stated above, both $\mathcal{I}$ and $\mathcal{I}'$ are isomorphic (to $\mathcal{Q}$). But that violates Corollary 4.8, as $\mathcal{I} \models \alpha$, while $\mathcal{I}' \models \neg\alpha$.

The claim implies, by PC, that either $\text{DL} \vdash \alpha$ or $\text{DL} \vdash \neg\alpha$, and hence one obtains the completeness of DL. $\square$

4.3 Categoricity

Now let us look at some very special first order theories.

Definition 4.3. A first order theory T is said to be *categorical*, if all its models are isomorphic to each other.

Using Proposition 4.5 and Corollary 4.8, it is easy to see that

Proposition 4.10. If T is categorical, it is complete.

Thus, any example of a categorical first order theory would also serve as an example of a complete theory.

Is there an example of a theory that is complete, but not categorical? Indeed, consider again the first order theory DL of Dense Linearly Ordered Sets without End-Points.

Proposition 4.11. DL is complete but not categorical.

Proof. It is already shown in Proposition 4.9 that DL is complete. Examples of models of DL are the sets $\mathcal{Q}$ and $\mathcal{R}$ of rational and real numbers respectively, indicating that DL is not categorical, as $\mathcal{Q}$ is countable, while $\mathcal{R}$ is not. $\square$

One also obtains

Proposition 4.12. If $\aleph$ is consistent, it cannot be categorical.